



Introduction

Genotype imputation is a common operation for diploid data. Although some approaches exist [2, 4], they are far less developed for autopolyploids, such as potato.

Here, we present our first results on **Imputato**, an approach that allows for efficient phasing and **imputation of haploids, diploids, and polyploids with no need for a pre-phased reference**, while such a reference can be utilized if available.

Core algorithm

The Imputato approach consists of initializing priors based on the original genotypes, with slight variations. For example, in a tetraploid, an allele count of 1 can translate into priors (0.24, 0.26, 0.23, 0.25). Similarly, a diploid allele count of 1 can translate into the priors (0.49, 0.50). These slight variations ensure that the different haploids are not completely identical.

In each iteration of Imputato, an independent Stephens-Li[6] style Hidden Markov Model (HMM) considers the priors for each haploid. These are used to express the haploid as a mosaic of other individuals and/or reference haplotypes. Since the data is haploid, the state space is linearly proportional to the number of other haplotypes, analogous to minimac[5].

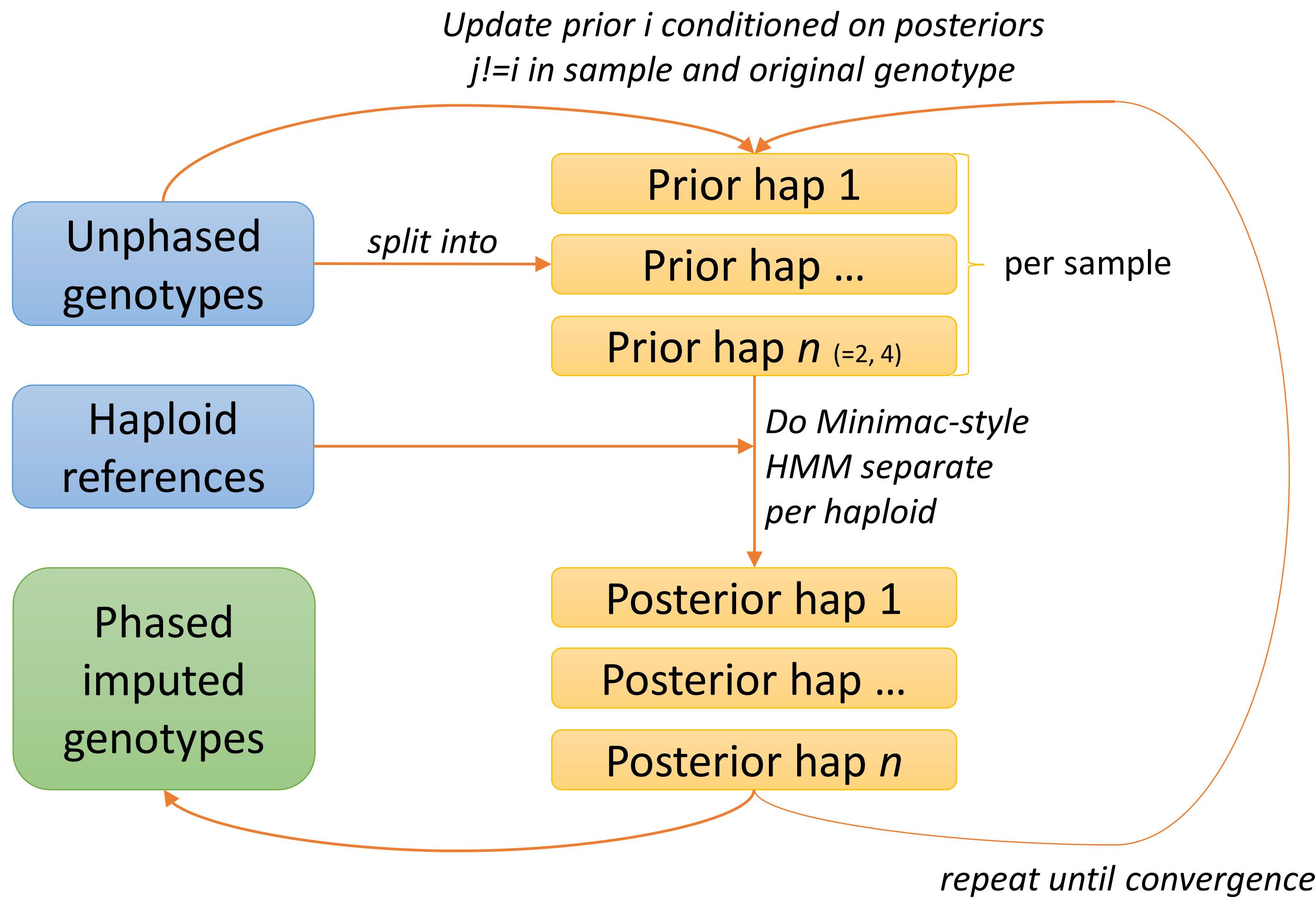


Figure 1. Overall algorithm. Unphased data and possible phased references are used. Priors are updated in successive iterations to ensure posteriors are not biased relative to known genotypes. At convergence, posteriors are used to impute.

Flips to untie the switch error knot

The forward-backward tables are collected for all HMM instances. These can be used to evaluate whether permuting the haploids of any sample downstream of some variant would result in a superior likelihood. If so, such a **flip** is executed. This approach is similar to one we have earlier shown to greatly reduce switch errors in family-based models[7]. Preliminary testing demonstrated that the flip handling yielded a moderate improvement to unphased imputed genotype concordance in Imputato as well.

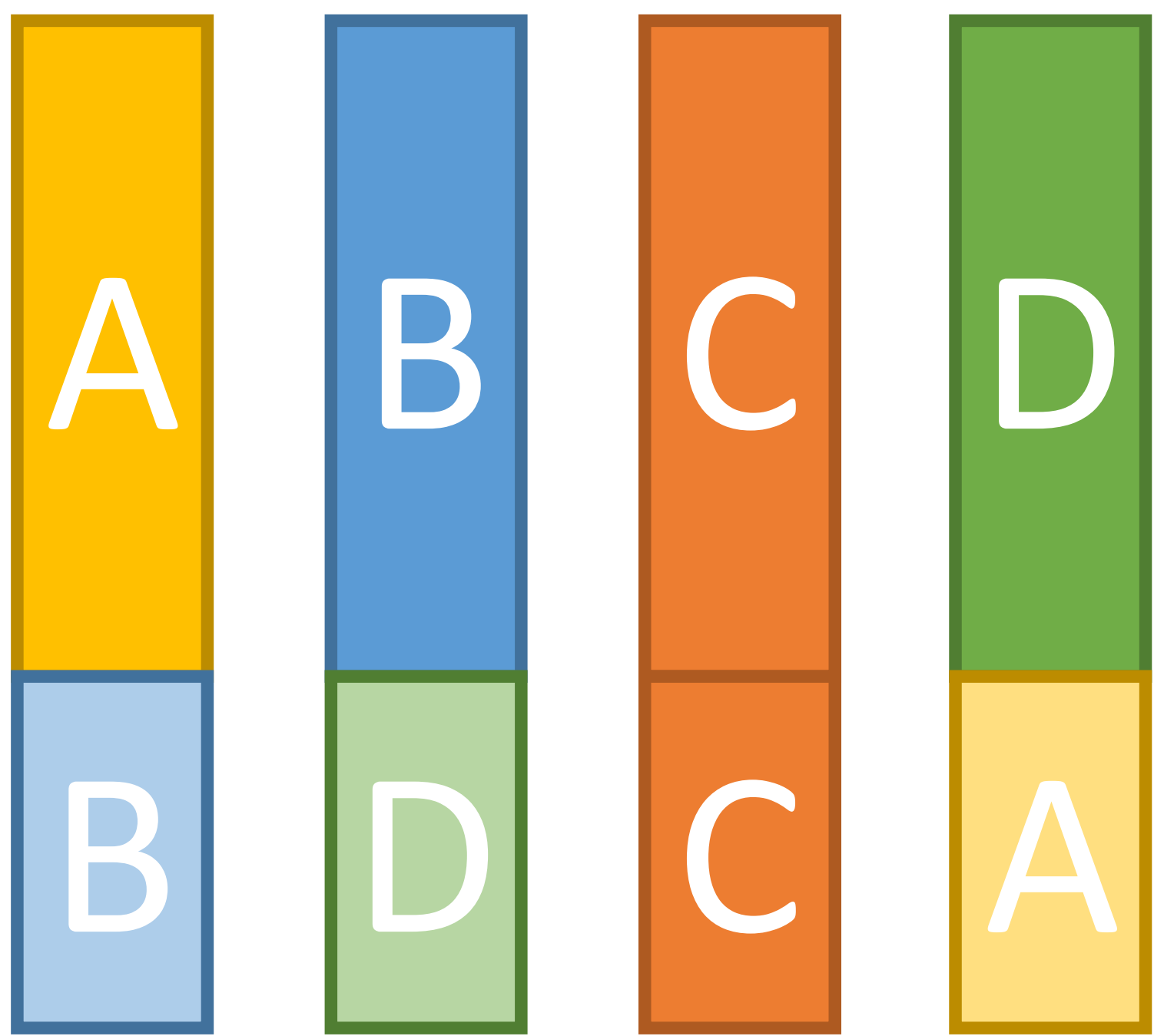


Figure 2. Illustration of how a flip can be detected and implemented in the four haploids constituting a single tetraploid individual sample. This increases accuracy. In this case, towards the end of the chromosome, the preliminary haplotypes ABCD are more optimal if the A is connected to the B tail, B to the D tail, C stays the same, and D connects to the A tail.

Nudging values

In a recent preprint [3], we used an iterative approach to reconstruct genotypes from pooling. Initial estimates of genotype likelihoods were produced for all individuals participating in a block of 16, distributed into 8 pools, with each individual in two pools. These likelihoods were fed to conventional, per-individual, imputation approaches.

Sometimes the combined genotype of all imputed samples would be inconsistent with the observed pooled data. This was addressed by increasing the priors for the missing alleles for all samples in the relevant pools, until results ended up consistent. That iterative feedback improved imputation accuracy, even for completely missing data.

Here, we use the same idea, considering each diploid or polyploid individual to be a pool of several, independent, haploids!

Datasets

In one scenario, we attempted to impute 240 random individuals from the 1000 Genomes[1] dataset, using the remaining individuals as phased references. We considered chromosome 20 and a scenario with the study individuals genotyped on Illumina Infinium OmniExpress-24, while the references were genotyped on Infinium Omni2.5-8, going from 17,015 to 52,697 SNPs.

For a tetraploid test case, we used published data from a breeding value evaluation experiment in potato[8]. In total, we used tetraploid genotypes for 895 SNPs from chromosome 1 in 669 tetraploid potatoes from a breeding program, which included full sibs to some extent. We artificially blanked out 50% of all genotypes at random and used the resulting dataset with no completely genotyped individuals and no pre-existing phasing as input data for Imputato.

Results

We focus on reporting concordance, that is, the deviation in dosage. We compute this across the full datasets, since we have different missingness patterns. That is, the LD genotypes in the human dataset, and the provided 50% genotypes in the potato dataset, are included in the overall numbers.

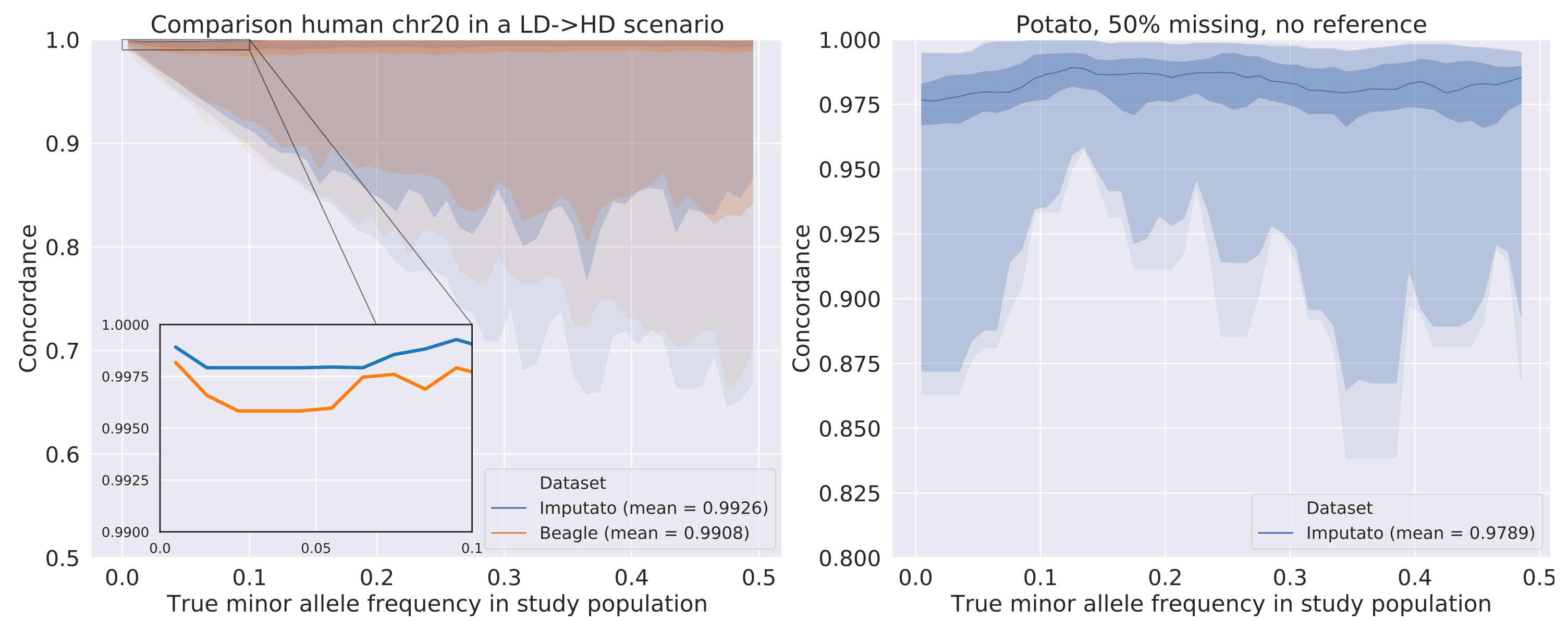


Figure 3. Results for: Left – the reference case of imputation from 17,015 low-density (LD) markers to 52,697 high-density (HD) markers using 1000 Genomes data on chromosome 20. Right – Internal imputation within 669 tetraploid potatoes for 895 SNPs on chromosome 1, with 50% of the individual genotypes blanked out at random. The solid lines indicate concordance in a sliding window. Shaded areas indicate various confidence bands, the tightest one being percentile 25-75, then 0.01-0.99 and outermost any single variant. The 25-75 band is tighter for Imputato on the human data, while it's slightly wider than Beagle in the distribution tails.

Imputato gives a higher average than Beagle, with a concordance exceeding 99% for our human data. The confidence bands are tighter down to 25% (the thinnest band in the top), when compared to Beagle, but the most extreme cases (percentile 1% and the final outliers) are slightly worse. For the potato data, the mean concordance is lower, and we have no reasonable baseline method to compare against. We can still conclude that we have fully acceptable results, and that the average concordance exceeds 97.5% across the full allele frequency spectrum.

Future work

The code base, found at <https://github.com/cnettel/imputato>, is still highly experimental and a work in progress, with different branches for different test cases. We hope to develop this into a stable and reusable tool.

Algorithm-wise, the most obvious future extension is to learn from other gradual optimization approaches, especially deep learning. Currently, Imputato takes slow steps in the posterior updates. Techniques such as mini-batches (using a subset of references each time), and momentum (accelerating changes that keep going in the same direction) could be directly borrowed from deep learning approaches and be applied here. This could bring improvements in speed as well as result quality.

We would also want to consider low-pass sequencing data rather than array genotyping. The lower results compared with Beagle for some variants might be attributed to numerical accuracy issues in the handling of the marker map, which we seek to address.

Furthermore, the very general nature of the prior-posterior feedback means that it would be possible to plug in other haploid imputation methods, as long as they can accept and produce genotype probabilities or likelihoods. That could imply more modern and efficient algorithms for vast references, or even combining family-based and reference-based imputation in a single feedback structure.

References

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